

1. General Description

The EM74HC74 and EM74HCT74 are dual positive edge triggered D-type flip-flop. They have individual data (nD), clock (nCP), set ($n\bar{S}D$) and reset ($n\bar{R}D$) inputs, and complementary nQ and $n\bar{Q}$ outputs. Data at the nD -input, that meets the set-up and hold time requirements on the LOW-to-HIGH clock transition, is stored in the flip-flop and appears at the nQ output. Schmitt-trigger action in the clock input, makes the circuit highly tolerant to slower clock rise and fall times. Inputs include clamp diodes that enable the use of current limiting resistors to interface inputs to voltages in excess of V_{CC} .

2. Features and Benefits

- Wide operating voltage 2.0 V to 6.0 V
- High noise immunity
- CMOS low power dissipation
- Input levels:
 - For EM74HC74: CMOS level
 - For EM74HCT74: TTL level
- Symmetrical output impedance
- Balanced propagation delays
- Latch-up performance exceeds 250 mA
- Complies with JEDEC standard:
 - JESD8C (2.7 V to 3.6 V)
 - JESD7A (2.0 V to 6.0 V)
- ESD protection:
 - HBM ANSI/ESDA/JEDEC JS-001 Class 3A exceeds 3500 V
 - CDM ANSI/ESDA/JEDEC JS-002 Class C3 exceeds 2000 V
- Multiple package options

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Dual D-type flip-flop with set and reset; positive edge-trigger

3. Ordering Information

Table 1. Ordering information

Type number	Topside marking	Package		
		Name	Description	Quantity
EM74HC74D	HC74□ YYYWW	SOP-14L	plastic small outline package; 14 leads; body width 3.9 mm	3000
EM74HCT74D	HCT74□ YYYWW			
EM74HC74PW	HC74□ YYYWW	TSSOP-14L	plastic thin shrink small outline package; 14 leads; body width 4.4 mm	3000
EM74HCT74PW	HCT74□ YYYWW			

MARKING INFORMATION

NOTE: HC74□ = Serial number and Internal code; YYYWW = Vendor Code and Date Code.

HC74 □

Internal Code
Serial number

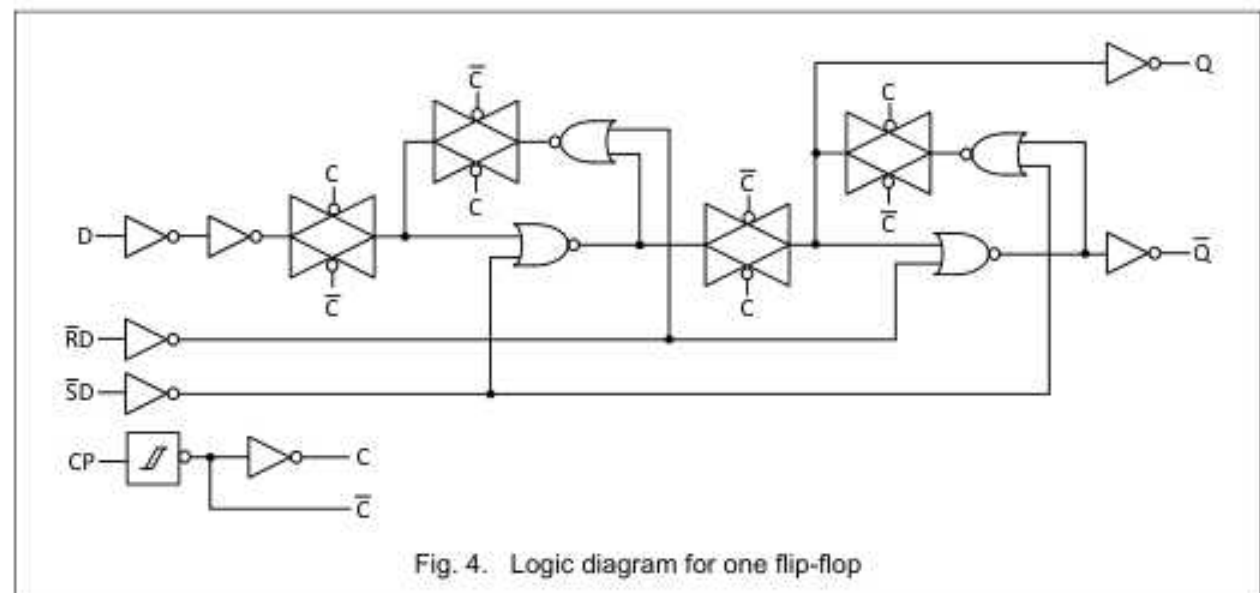
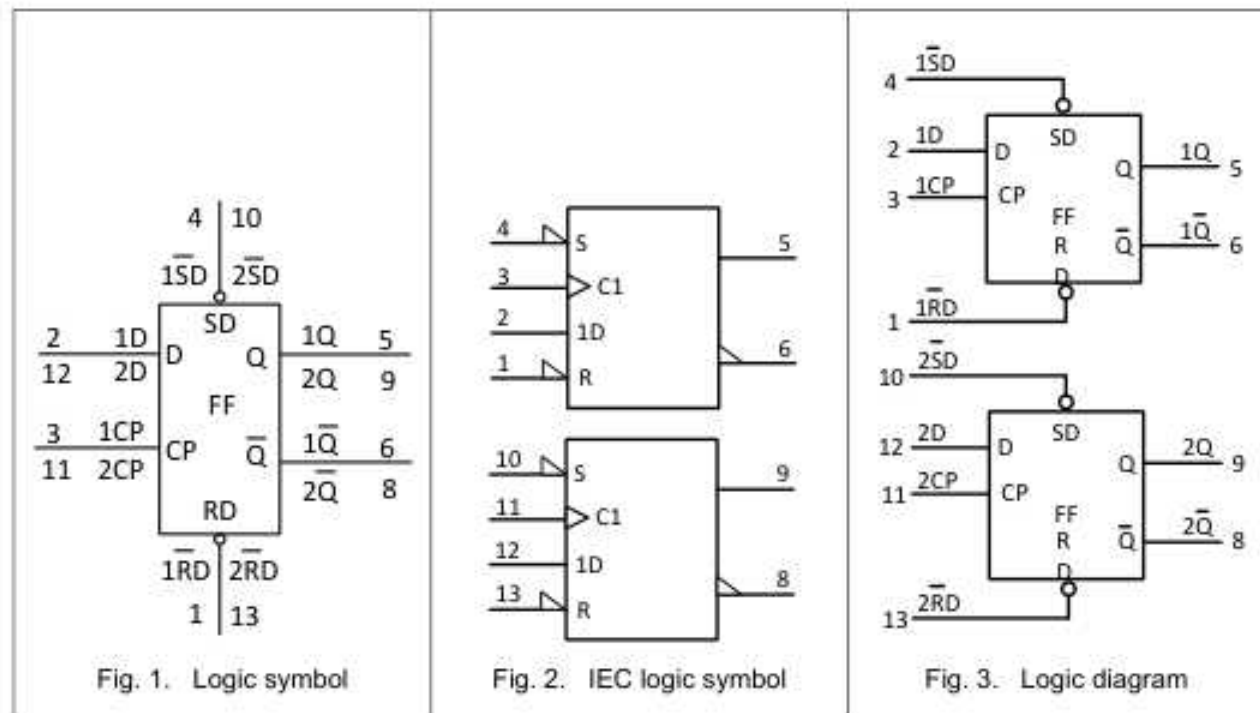
X YY WW

Date Code -Week
Date Code -Year
Vendor Code

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4. Function Diagram



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5. Pinning Information

5.1. Pinning

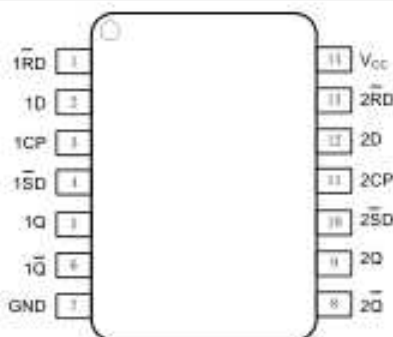


Fig. 5. Top view pin configuration SOP-14L and TSSOP14L

5.2. Pin description

Table 2. Pin description

Symbol	Pin	Description
1RD, 2RD	1, 13	asynchronous reset-direct input (active LOW)
1D, 2D	2, 12	data output
1CP, 2CP	3, 11	clock input (LOW-to-HIGH, edge-triggered)
1SD, 2SD	4, 10	asynchronous set-direct input (active LOW)
1Q, 2Q	5, 9	output
1Q, 2Q	6, 8	complement output
GND	7	ground (0 V)
Vcc	14	supply voltage

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6. Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level; X = don't care.

Input				Output	
$\overline{nSD}$	$\overline{nRD}$	nCP	nD	nQ	$\overline{nQ}$
L	H	X	X	H	L
H	L	X	X	L	H
L	L	X	X	H	H

Table 4. Function table

H = HIGH voltage level; L = LOW voltage level; X = don't care;

↑ = LOW-to-HIGH transition; Q_{n+1} = state after the next LOW-to-HIGH CP transition.

Input				Output	
$\overline{nSD}$	$\overline{nRD}$	nCP	nD	Q_{n+1}	$\overline{Q}_{n+1}$
H	H	↑	L	L	H
H	H	↑	H	H	L

7. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

Table 5. Absolute Maximum Ratings

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

Symbol	Parameter	Conditions	Min	Max	Unit
V_{CC}	supply voltage		-0.5	7	V
I_{IK}	input clamping current	$V_I < -0.5 \text{ V}$ or $V_I > V_{CC} + 0.5 \text{ V}$		±20	mA
I_{OK}	output clamping current	$V_O < -0.5 \text{ V}$ or $V_O > V_{CC} + 0.5 \text{ V}$		±20	mA
I_O	output current	$-0.5 \text{ V} < V_O < V_{CC} + 0.5 \text{ V}$		±25	mA
I_{CC}	supply current			100	mA
I_{GND}	ground current		-100		mA
P_{tot}	total power dissipation	$T_{amb} = -40 \text{ }^\circ\text{C}$ to $+125 \text{ }^\circ\text{C}$		500	mW
T_{stg}	storage temperature		-65	+150	$^\circ\text{C}$

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8. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

Table 6. Recommended Operating Conditions

Symbol	Parameter	Conditions	EM74HC74			EM74HCT74			Unit
			Min	Typ	Max	Min	Typ	Max	
V_{CC}	supply voltage		2.0	5.0	6.0	4.5	5.0	5.5	V
V_I	input voltage		0		V_{CC}	0		V_{CC}	V
V_O	output voltage		0		V_{CC}	0		V_{CC}	V
T_{amb}	ambient temperature		-40	+25	+125	-40	+25	+125	°C
$\Delta t/\Delta V$	input transition rise and fall rate	$V_{CC} = 2.0\text{ V}$			625				ns/V
		$V_{CC} = 4.5\text{ V}$		1.67	139		1.67	139	ns/V
		$V_{CC} = 6.0\text{ V}$			83				ns/V

9. Static Characteristics

Table 7. Static characteristics

At recommended operating conditions. Voltages are referenced to GND (ground = 0 V).

Recommended operating conditions: Voltages are referenced to GND (V _{Ground} = 0 V).								
Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
EM74HC74								
V _{IH}	HIGH-level input voltage	V _{CC} = 2.0 V	1.5	1.2		1.5		V
		V _{CC} = 4.5 V	3.15	2.5		3.15		V
		V _{CC} = 6.0 V	4.2	3.3		4.2		V
V _{IL}	LOW-level input voltage	V _{CC} = 2.0 V		0.8	0.5		0.5	V
		V _{CC} = 4.5 V		2.1	1.35		1.35	V
		V _{CC} = 6.0 V		2.8	1.8		1.8	V
V _{OH}	HIGH-level output voltage	V _I = V _{IH} or V _{IL}						
		I _O = -4.0 mA; V _{CC} = 4.5 V	3.84	4.4		3.7		V
		I _O = -5.2 mA; V _{CC} = 6.0 V	5.34	5.9		5.2		V

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Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
V _{OL}	LOW-level output voltage	V _I = V _{IH} or V _{IL}						
		I _O = 4.0 mA; V _{CC} = 4.5 V		0.03	0.33		0.4	V
		I _O = 5.2 mA; V _{CC} = 6.0 V		0.04	0.33		0.4	V
I _I	Input leakage current	V _I = V _{CC} or GND ; V _{CC} = 6.0 V		0.01	±1		±1	µA
I _{CC}	supply current	V _I = V _{CC} or GND ; I _O = 0A ; V _{CC} = 6.0 V		0.01	20		40	µA
C _I	input capacitance			7				pF
EM74HCT74								
V _{IH}	HIGH-level input voltage	V _{CC} = 4.5 V to 5.5 V	2.0	1.7		2.0		V
V _{IL}	LOW-level input voltage	V _{CC} = 4.5 V to 5.5 V		1.3	0.8		0.8	V
V _{OH}	HIGH-level output voltage	V _I = V _{IH} or V _{IL} ; V _{CC} = 4.5 V						
		I _O = -4.0 mA	3.84	4.4	-	3.7		V
V _{OL}	LOW-level output voltage	V _I = V _{IH} or V _{IL} ; V _{CC} = 4.5 V						
		I _O = 4.0 mA		0.03	0.33		0.4	V
I _I	Input leakage current	V _I = V _{CC} or GND ; V _{CC} = 5.5 V		0.01	±1		±1	µA
I _{CC}	supply current	V _I = V _{CC} or GND ; I _O = 0A ; V _{CC} = 5.5 V		0.01	20		40	µA
ΔI _{CC}	additional supply current	V _I = V _{CC} - 2.1 V; I _O = 0 A; other inputs at V _{CC} or GND; V _{CC} = 4.5 V to 5.5 V						
		per input pin; nD, nRD inputs		350	500		550	µA
		per input pin; nSD inputs		640	850		950	µA
		per input pin; nCP inputs		1130	1550		1650	µA
C _I	input capacitance			7				pF

[1] All typical values are measured at T_{amb} = 25°C.

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10. Dynamic Characteristics

Table 8. Dynamic characteristics

Voltages are referenced to GND (ground = 0 V); for test circuit see Fig. 8.

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
EM74HC74								
t _{pd}	propagation delay	nCP to nQ, nQ̄; see Fig. 6 [2]						
		V _{CC} = 2.0 V			30		33	ns
		V _{CC} = 4.5 V			16		19	ns
		V _{CC} = 6.0 V			12		14	ns
		nSD̄ to nQ, nQ̄; see Fig. 7 [2]						
		V _{CC} = 2.0 V			30		33	ns
		V _{CC} = 4.5 V			16		19	ns
		V _{CC} = 6.0 V			12		14	ns
		nRD̄ to nQ, nQ̄; see Fig. 7 [2]						
		V _{CC} = 2.0 V			30		33	ns
		V _{CC} = 4.5 V			16		19	ns
		V _{CC} = 6.0 V			12		14	ns
t _t	transition time	nQ, nQ̄; see Fig. 6 [3]						
		V _{CC} = 2.0 V			9		11	ns
		V _{CC} = 4.5 V			6		8	ns
		V _{CC} = 6.0 V			4		5	ns
t _w	pulse width	nCP HIGH or LOW; see Fig. 6						
		V _{CC} = 2.0 V	100			120		ns
		V _{CC} = 4.5 V	20			24		ns
		V _{CC} = 6.0 V	17			20		ns
		nSD̄, nRD̄ LOW; see Fig. 7						
		V _{CC} = 2.0 V	100			120		ns
		V _{CC} = 4.5 V	20			24		ns
		V _{CC} = 6.0 V	17			20		ns
t _{rec}	recovery time	nSD̄, nRD̄; see Fig. 7						
		V _{CC} = 2.0 V	40			45		ns
		V _{CC} = 4.5 V	8			9		ns

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Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
		$V_{CC} = 6.0 \text{ V}$	7			8		ns
t_{su}	set-up time	nD to nCP; see Fig. 6						
		$V_{CC} = 2.0 \text{ V}$	75			90		ns
		$V_{CC} = 4.5 \text{ V}$	15			18		ns
		$V_{CC} = 6.0 \text{ V}$	13			15		ns
t_h	hold time	nD to nCP; see Fig. 6						
		$V_{CC} = 2.0 \text{ V}$	3			3		ns
		$V_{CC} = 4.5 \text{ V}$	3			3		ns
		$V_{CC} = 6.0 \text{ V}$	3			3		ns
f_{max}	maximum frequency	nCP; see Fig. 6						
		$V_{CC} = 2.0 \text{ V}$	4.8			4.0		MHz
		$V_{CC} = 4.5 \text{ V}$	24			20		MHz
		$V_{CC} = 6.0 \text{ V}$	28			24		MHz
C_{PD}	power dissipation capacitance	$V_I = \text{GND to } V_{CC};$ $f = 1\text{MHz}$	[4]	23				pF
EM74HCT74								
t_{pd}	propagation delay	nCP to nQ, n $\bar{Q}$; see Fig. 6 [2]						
		$V_{CC} = 4.5 \text{ V}$			16		19	ns
		n $\bar{S}$ D to nQ, n $\bar{Q}$; see Fig. 7 [2]						
		$V_{CC} = 4.5 \text{ V}$			16		19	ns
		n $\bar{R}$ D to nQ, n $\bar{Q}$; see Fig. 7 [2]						
		$V_{CC} = 4.5 \text{ V}$			16		19	ns
t_z	transition time	nQ, n $\bar{Q}$; see Fig. 6 [3]						
		$V_{CC} = 4.5 \text{ V}$			6		8	ns
t_w	pulse width	nCP HIGH or LOW; see Fig. 6						
		$V_{CC} = 4.5 \text{ V}$	23			27		ns
		n $\bar{S}$ D, n $\bar{R}$ D LOW; see Fig. 7						
		$V_{CC} = 4.5 \text{ V}$	20			24		ns
t_{rec}	recovery time	n $\bar{S}$ D, n $\bar{R}$ D; see Fig. 7						
		$V_{CC} = 4.5 \text{ V}$	8			9		ns

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Dual D-type flip-flop with set and reset; positive edge-trigger

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
t_{su}	set-up time	nD to nCP; see Fig. 6						
		$V_{CC} = 4.5\text{ V}$	15			18		ns
t_{th}	hold time	nD to nCP; see Fig. 6						
		$V_{CC} = 4.5\text{ V}$	3			3		ns
f_{max}	maximum frequency	nCP; see Fig. 6						
		$V_{CC} = 4.5\text{ V}$	22			18		MHz
C_{PD}	power dissipation capacitance	$V_I = \text{GND to } V_{CC};$ $f = 1\text{MHz}$ [4]		24				pF

[1] Typical values are measured at $T_{amb} = 25\text{ °C}$.

[2] t_{dL} is the same as t_{dLH} and t_{dHL} .

[3] t_i is the same as t_{iLH} and t_{iHL} .

[4] C_{PD} is used to determine the dynamic power dissipation (P_D in μW):

$P_D = C_{PD} \times V_{CC}^2 \times f_i \times N + \sum(C_L \times V_{CC}^2 \times f_o)$ where:

f_i = input frequency in MHz;

f_o = output frequency in MHz;

C_L = output load capacitance in pF;

V_{CC} = supply voltage in V;

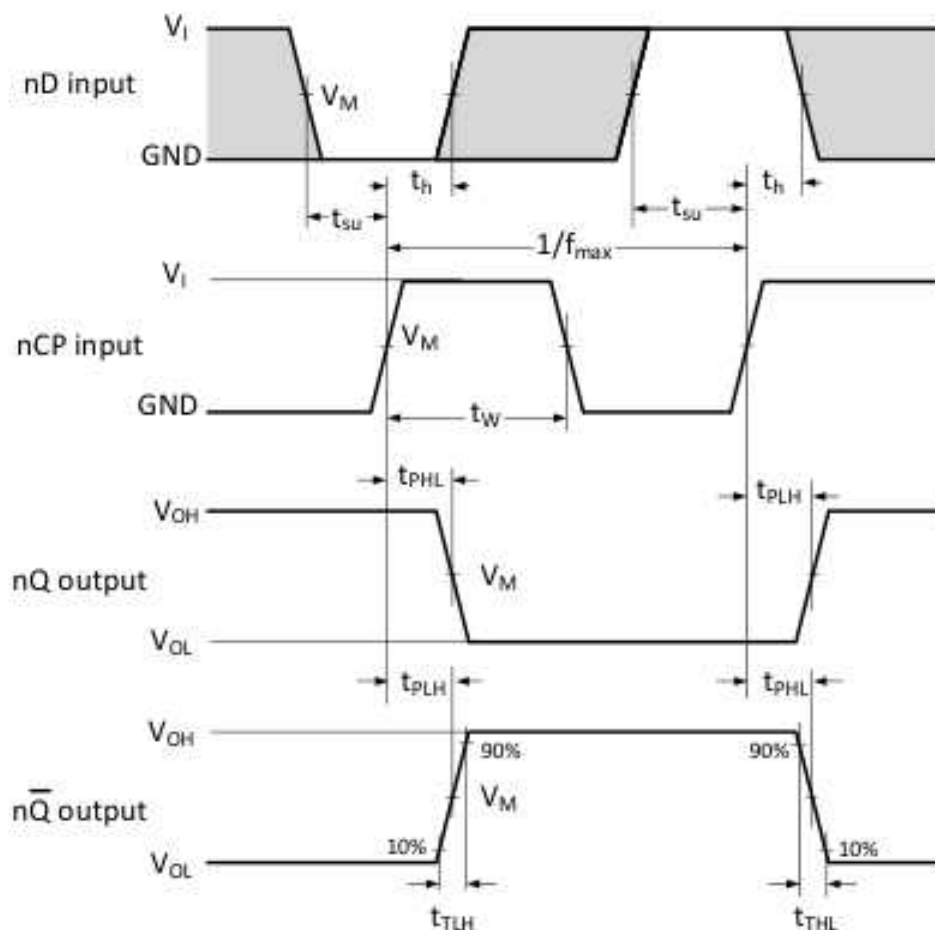
N = number of inputs switching;

$\sum(C_L \times V_{CC}^2 \times f_o)$ = sum of outputs.

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10.1. Waveforms and test circuit



Measurement points are given in Table 9.

V_{OL} and V_{OH} are typical output voltage levels that occur with the output load.

Fig. 6. The Propagation delay input (CP) to output (Qn), output transition time, clock input (CP) pulse width and the maximum frequency (CP)

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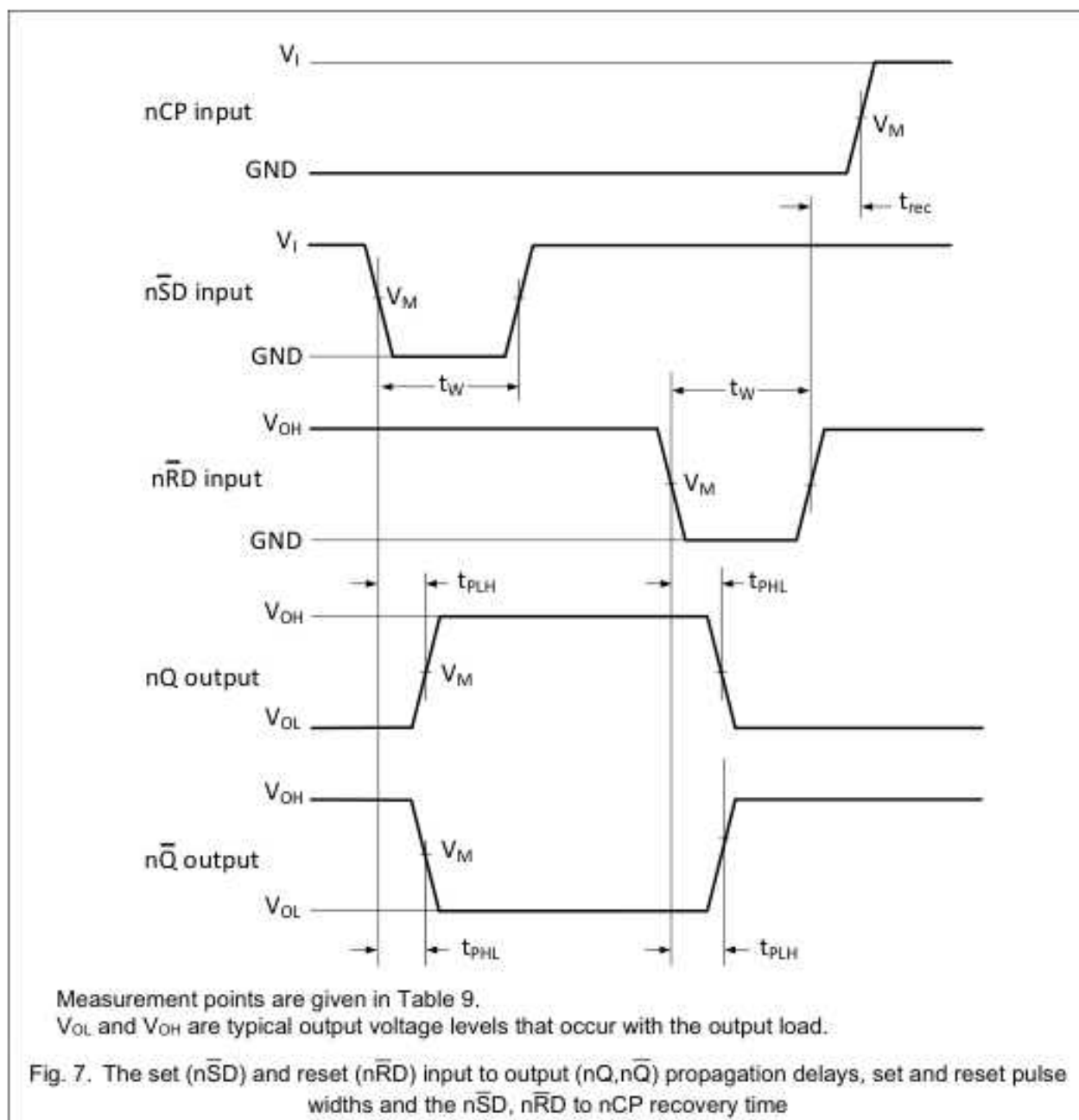
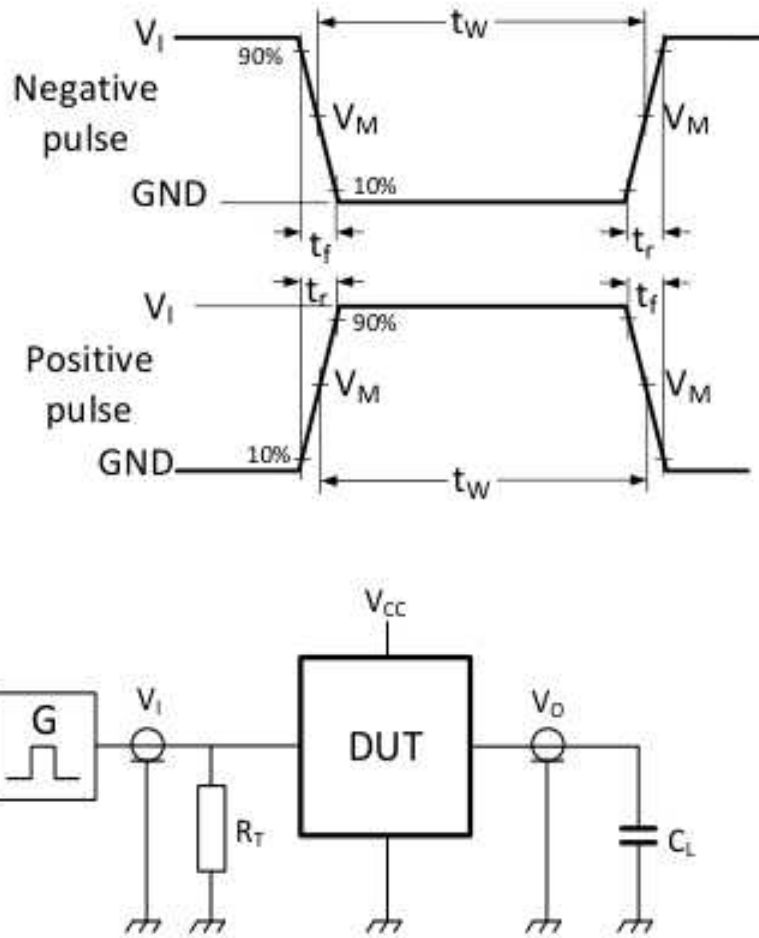


Table 9. Measurement points

Type	Input	Output
	V_M	V_M
EM74HC74	$0.5V_{CC}$	$0.5V_{CC}$
EM74HCT74	1.3V	1.3V

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Test data is given in Table 10.

Definitions for test circuit:

R_L = Load resistance.

C_L = Load capacitance including jig and probe capacitance.

R_T = Termination resistance should be equal to the output impedance Z_o of the pulse generator..

Fig. 8. Test circuit for measuring switching times

Table 10. Test data

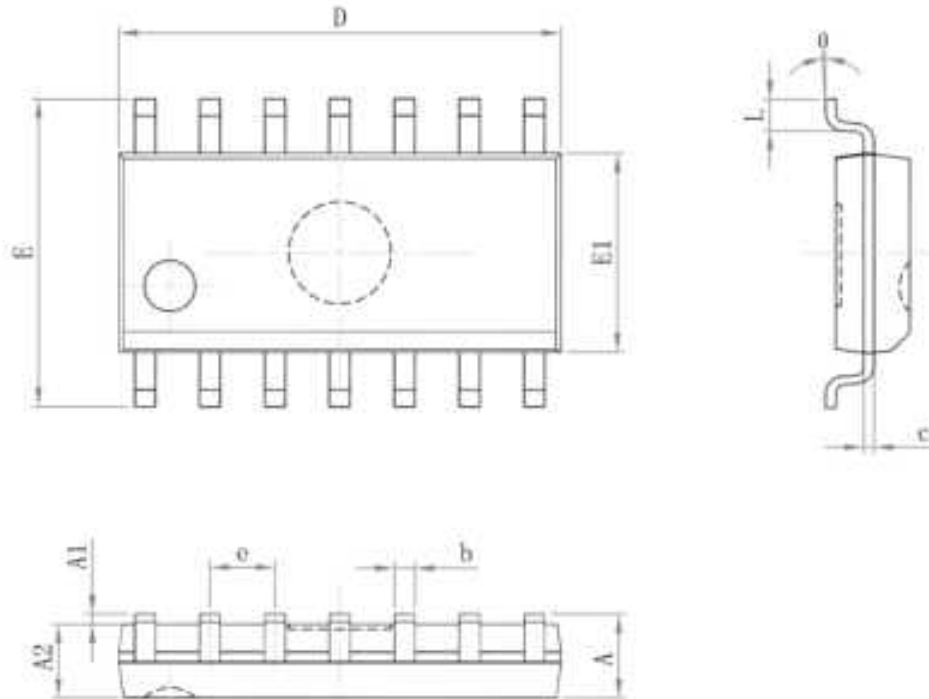
Type	Input		Load	Test
	V_I	$t_r = t_f$	C_L	
EM74HC74	V_{CC}	≤ 2.5 ns	15 pF	t_{PLH} , t_{PHL}
EM74HCT74	3 V	≤ 2.5 ns	15 pF	t_{PLH} , t_{PHL}

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11. Package Outline

SOP-14L

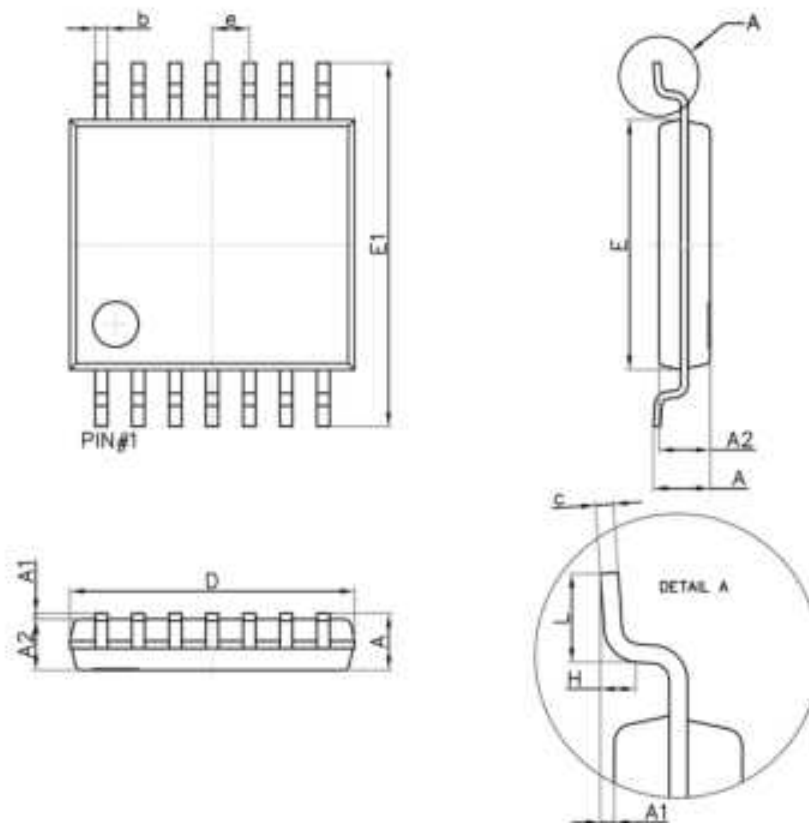


Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min	Max	Min	Max
A	—	1.750	—	0.069
A1	0.100	0.250	0.004	0.010
A2	1.250	—	0.049	—
b	0.310	0.510	0.012	0.020
c	0.100	0.250	0.004	0.010
D	8.450	8.850	0.333	0.348
E	5.800	6.200	0.228	0.244
E1	3.800	4.000	0.150	0.157
e	1.270(BSC)		0.050(BSC)	
L	0.400	1.270	0.016	0.050
θ	0°	8°	0°	8°

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TSSOP-14L



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
D	4.900	5.100	0.193	0.201
E	4.300	4.500	0.169	0.177
b	0.190	0.300	0.007	0.012
c	0.090	0.200	0.004	0.008
E1	6.250	6.550	0.246	0.258
A	—	1.200	—	0.047
A2	0.800	1.000	0.031	0.039
A1	0.050	0.150	0.002	0.006
e	0.65 (BSC)		0.026 (BSC)	
L	0.500	0.700	0.020	0.028
H	0.25(TYP)		0.01(TYP)	
θ	1°	7°	1°	7°

12. Tape and Reel Information

12.1. Carrier tape dimensions

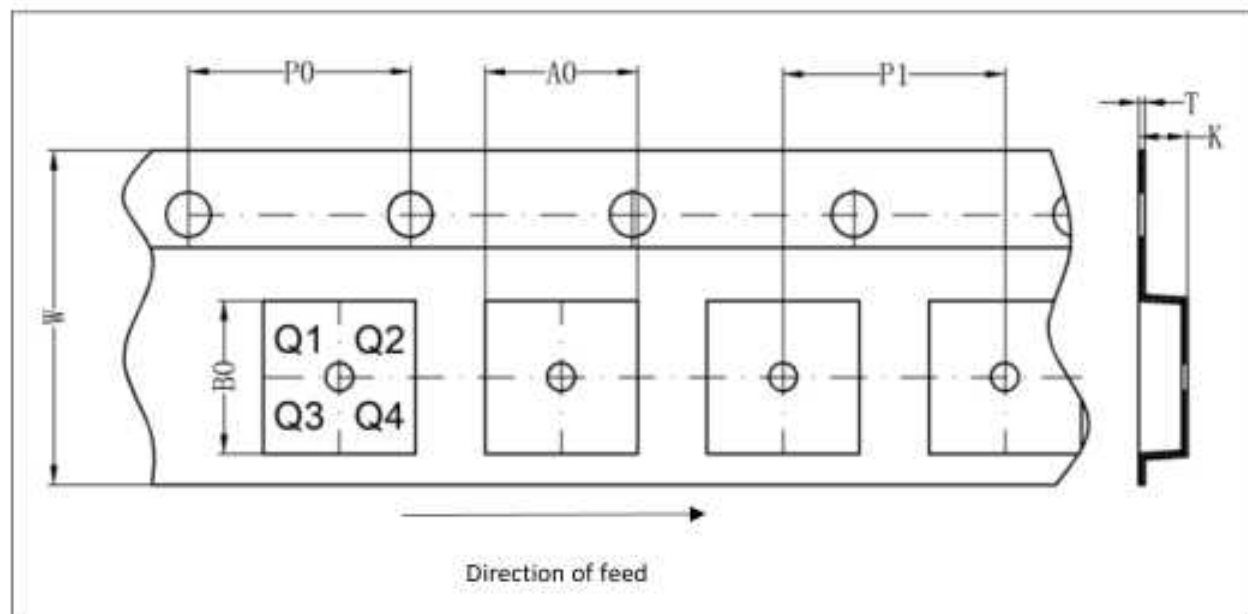


Table 11. Carrier tape dimensions

Package version	A0(mm)	B0(mm)	K0(mm)	T(mm)	P1(mm)	W(mm)	P0(mm)	PIN 1
SOP-14L	6.35	9.1	1.9	0.22	8	16	4	Q1
TSSOP-14L	6.7	5.45	1.6	0.25	8	12	4	Q1

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12.2. Reel and box dimensions

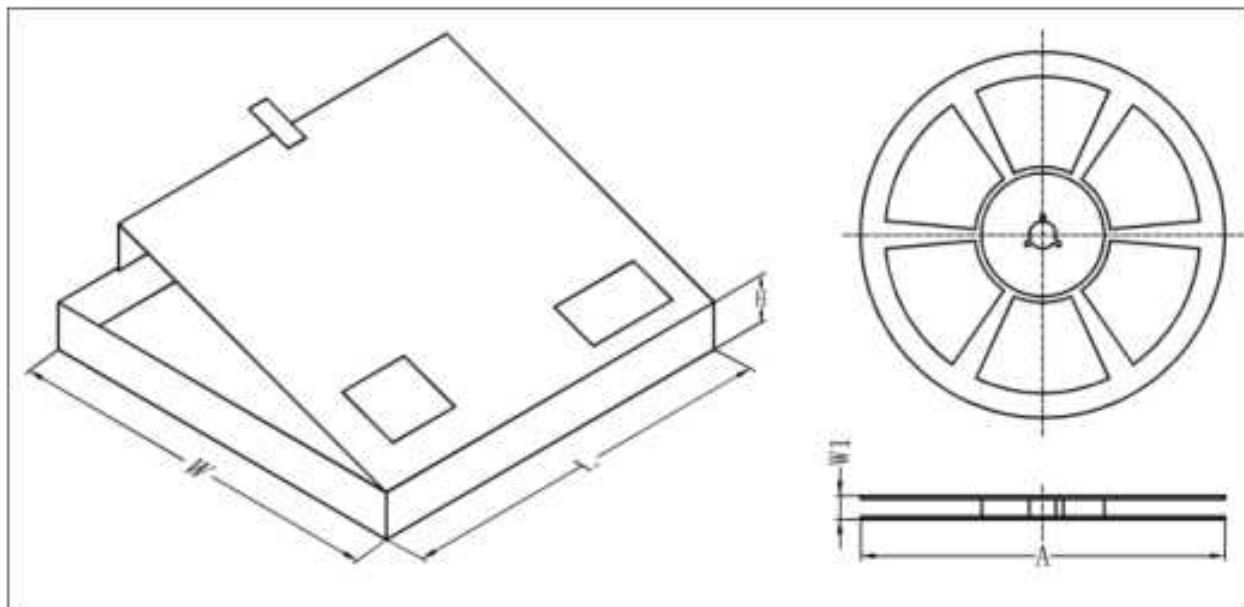


Table 12. Dimensions and quantities

Package version	Type NO. ending	Reel Dimension A (mm)	Reel Width W1 (mm)	SPQ (pcs) [1]	Reels per box	Outer box dimensions L×W×H(mm) [2]
SOP-14L	D	330	22.4	3000	1	358x340x50
TSSOP-14L	PW	330	18.4	3000	1	358x340x50

[1] Packing quantity dependent on specific product type. Please contact your local Energymath representative for ordering.

[2] Dimensions for reference only.

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13. Abbreviations

Table 13. Abbreviations

Acronym	Description
CMOS	Complementary Metal-Oxide Semiconductor
DUT	Device Under Test
ESD	ElectroStatic Discharge
HBM	Human Body Model
CDM	Charged Device Model
TTL	Transistor-Transistor Logic

14. Revision History

Table 14. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EM74HC_HCT74 Rev. 1.2	Mar 17, 2026	Product datasheet		EM74HC_HCT74 Rev. 1.1
Modifications:	Section 12 added Tape and Reel Information.			
EM74HC_HCT74 Rev. 1.1	Dec 15, 2025	Product datasheet		EM74HC_HCT74 Rev. 1.0
Modifications:	Table 1: topside marking information updated.			
EM74HC74_HCT74 Rev. 1.0	Aug 28, 2024	Product datasheet		